PEDRO FREDERICO SILVA

SOFTWARE DEVELOPER

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SUMMARY

Backend and systems engineer with 7+ years of experience building scalable distributed systems and internal tooling in high-scale production environments. Currently working at Euronext Securities, developing mission-critical event-driven services using Java, Kafka, and cloud-native architectures in containerized OpenShift environments.
Strong focus on system reliability, observability, and developer productivity tooling, with hands-on experience building internal platforms, CI/CD pipelines, and service integrations across complex enterprise ecosystems.
Experienced in designing and operating distributed systems with emphasis on performance, fault tolerance, and operational visibility. Passionate about building tools that improve engineering workflows and enable teams to ship software reliably at scale. Open to remote opportunities and international relocation within backend, platform engineering, and developer tooling environments.

SKILLS

Backend & Distributed Systems: Java, Spring Boot, Quarkus, REST APIs, Kafka, event-driven architecture, microservices, SOAP, Oracle SQL, distributed systems design, system design
Platform / DevOps / Infrastructure: Docker, Jenkins, OpenShift, GitLab, Git, Bitbucket, CI/CD pipelines, observability, OpenTelemetry, Prometheus, metrics & tracing, cloud systems (AWS)
Tooling & Internal Systems: Internal developer tools, workflow automation, service integration, API design, platform engineering concepts
Frontend (supporting tools): React, Next.js, TypeScript, JavaScript, HTML, CSS
Other: Data Structures, Big-O, Python, C++, Unity (bonus), C#, CMake

WORK EXPERIENCE

BACKEND / PLATFORM ENGINEER

(2026 - Present)

Euronext Securities

- Developed and maintained mission-critical distributed backend services in a high-throughput financial post-trade system using Java, Quarkus, and Kafka
- Designed and implemented event-driven workflows using the Outbox Pattern to ensure data consistency and reliable message processing across services
- Contributed to the evolution of internal service architecture in a containerized OpenShift environment, supporting scalable deployment and operations of microservices
- Implemented observability across distributed services using OpenTelemetry and Prometheus, enabling end-to-end tracing, metrics collection, and improved incident diagnosis
- Built an internal developer productivity tool integrating GitHub Copilot with Jira and Confluence through the Model Context Protocol (MCP), leveraging LLM-based context retrieval to assist with user story execution and code generation.
- Collaborated with cross-functional engineering teams to improve service reliability, deployment workflows, and system visibility across distributed components.

FULL STACK / SYSTEMS ENGINEER

(2019 - 2025)

Vodafone (Celfocus)

- Developed and maintained large-scale distributed systems for telecom clients (Vodafone international projects), serving millions of users across multiple regions
- Designed and implemented backend services using Java, Spring Boot, and microservices architectures, supporting scalable and resilient system operations
- Led migration initiatives from monolithic architectures to microservices, improving deployment independence and system scalability
- Integrated asynchronous processing systems using AWS SQS, improving throughput and decoupling of backend services
- Contributed to CI/CD pipeline design and maintenance using Jenkins and containerized deployment strategies with Docker
- Supported system observability and production troubleshooting across distributed environments using logs, metrics, and system-level diagnostics
- Mentored junior engineers and contributed to onboarding documentation and engineering best practices within the team
- Participated in cross-team collaboration and technical alignment in Agile environments (Scrum ceremonies, backlog refinement, incident resolution)

GIS TECHNICIAN / SURVEYING ENGINEERING (2017 - 2018)

Eri Engenharia / Viamapa

- Performed geographic data analysis and digital mapping for infrastructure projects using ArcGIS.

EDUCATION AND CERTIFICATION

BACHELOR'S IN GAME DEVELOPMENT ENGINEERING (2021 - 2024)
Instituto Politécnico do Cávado e do Ave

MASTER'S IN GEOSPATIAL ENGINEERING (2015 - 2017)
Faculdade de Ciências da Universidade do Porto

BACHELOR'S IN GEOSPATIAL ENGINEERING (2010 - 2015)
Faculdade de Ciências da Universidade do Porto

INTENSIVE PROGRAMMING BOOTCAMP (2018)
Academia de Código | 2018

AWS CERTIFIED CLOUD PRACTITIONER (2025)

CORE POSITIONONG

- Distributed systems & backend scalability
- Internal developer tooling & automation
- Observability & platform reliability engineering
- CI/CD and cloud-native infrastructure
- Workflow optimization for engineering teams